

Project Development Phase

Model Performance Test

Project Name	AI-Enhanced Intrusion Detection System
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Model Performance Testing:

Project team has filled the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	Classification Model: • Confusion Matrix – Generated via Scikit-Learn on test split • Accuracy Score – 98.9% • Classification Report – Precision/Recall/F1 per class (BENIGN, Brute Force, SQL Injection, XSS) Regression Model: N/A (Classification task)	
2.	Tune the Model	Hyperparameter Tuning – RandomForestClassifier optimized: n_estimators=100, max_depth=None, min_samples_split=2 Validation Method – Train/Test Split (80/20) with SMOTE applied to training fold only	